COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE 21S3
DATE PERFORMED:	10/13/2026	**DATE SUBMITTED:**	10/13/26
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ACTIVITY NO. 12 **Algorithmic Strategies**

1. PROCEDURE OUTPUT

Strategy	Algorithm	Analysis
Recursion	Factorial Calculation	Breaks down a problem into a base case and recursive calls where a function solves smaller subproblems
Brute Force	Linear Search	Tests every possible solution until the correct match is found
Backtracking	Sudoku Solver	Builds a grid of solutions and backtracks as soon as a cell violates the grid restriction
Greedy	Shortest Path	Makes the locally optimal choice at each step hoping to reach the global optimum. (Does not guarantee global optimality for all problems)
Divide-and-Conquer	Merge Sort	Divides a problem into smaller independent subproblems, solves each one, then combines their solutions

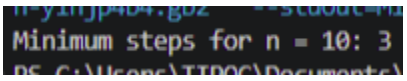
Screenshot	
Analysis	The code uses recursion to find the minimum number of steps needed to reduce a number to 1. It saves the answers in the memo array so it would not need to solve the same problem again. It also checks if the number can be divided by 2 or 3 and chooses the smallest number of steps. This makes the program run faster and avoid repeating the same calculations.

Table 12-2. Memoization Implementation

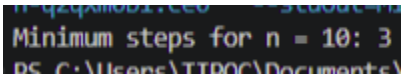
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Analysis	This code starts from 1 and calculates the minimum steps for each number until it reaches the given number. It stores the result in the dp array and compares the possible operations to get the smallest answer. Unlike memoization, it uses a loop instead of recursion, which makes the process more organized because it solves the smaller problems first.

Table 12-3. Bottom-Up Dynamic Programming Implementation

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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